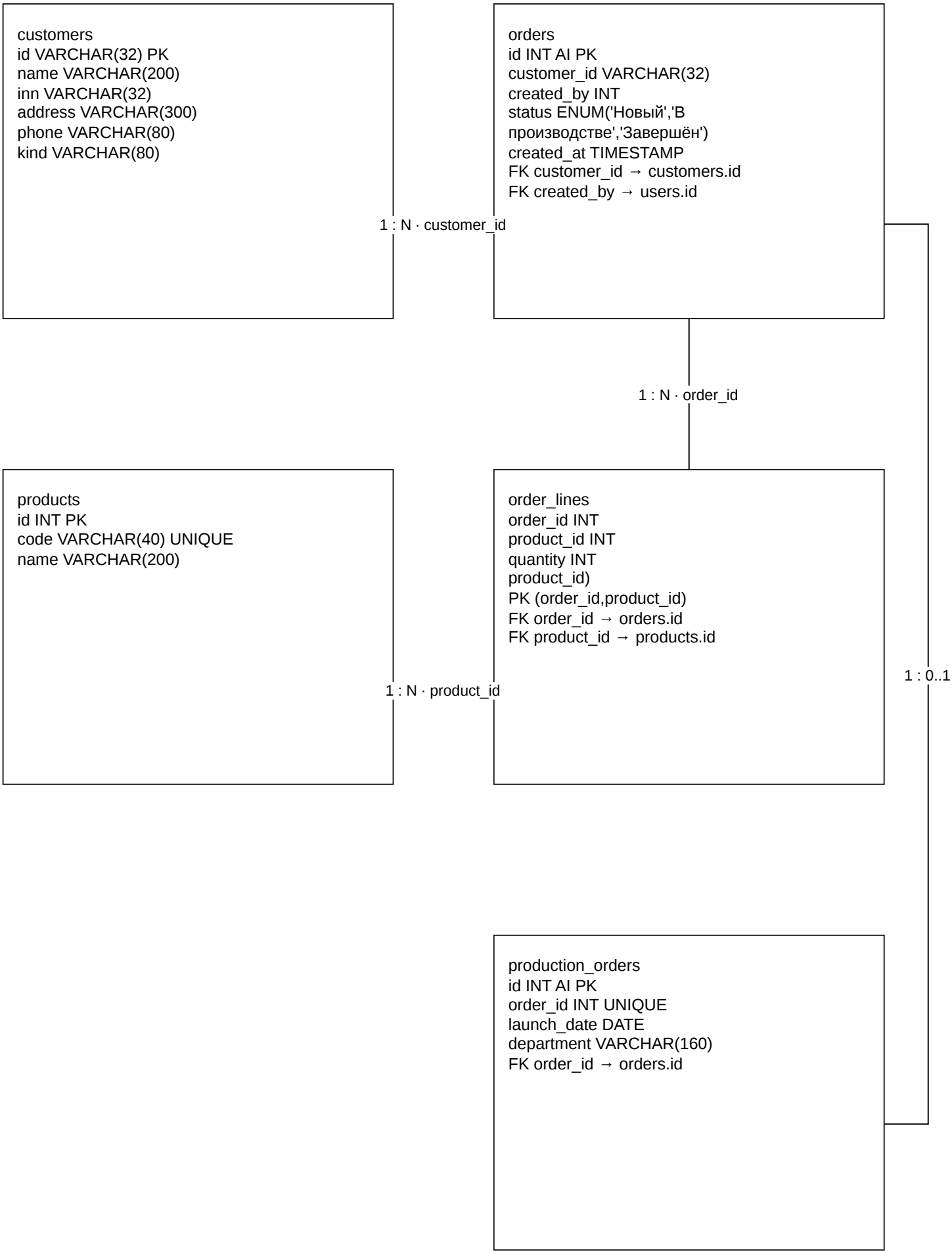
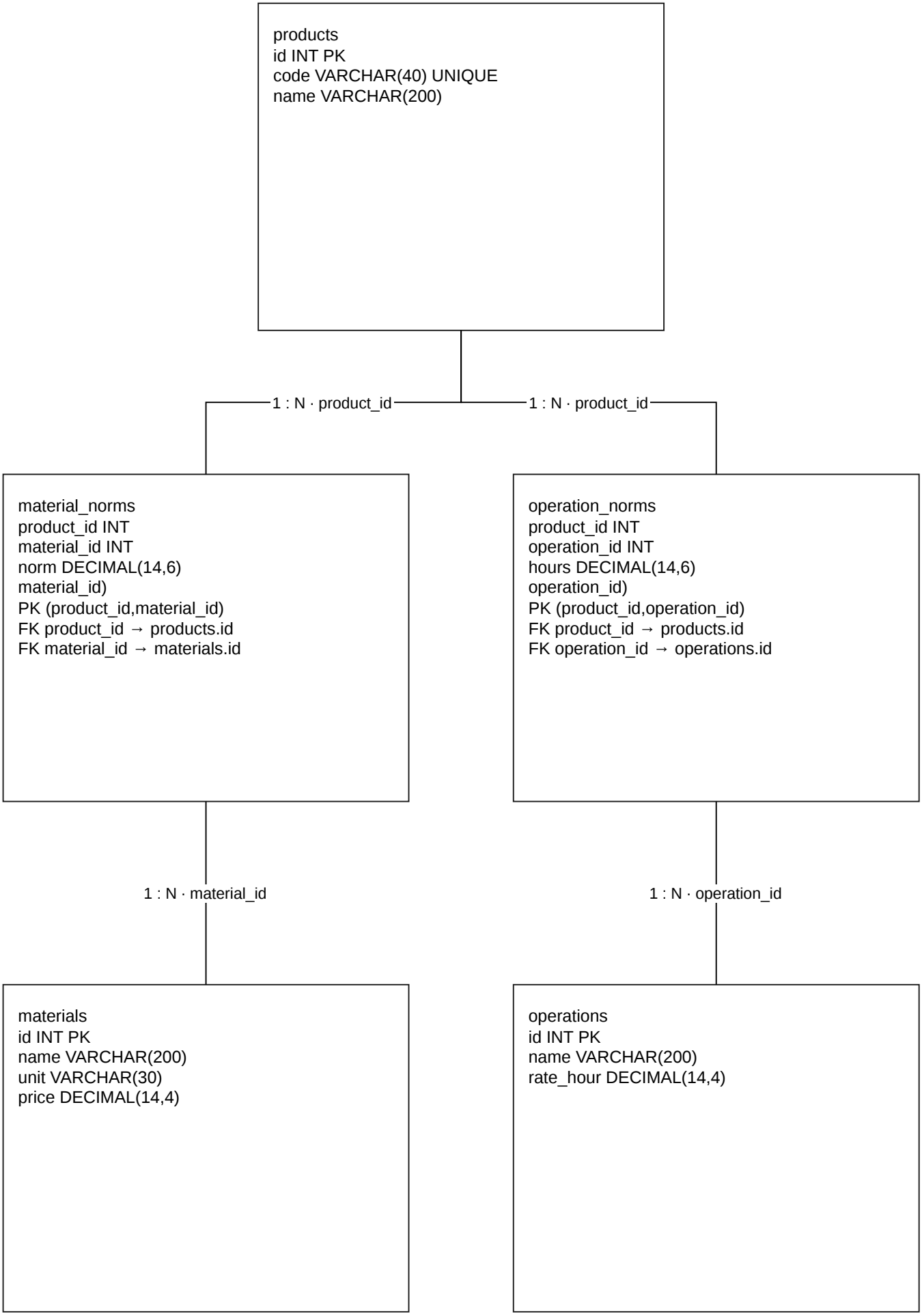
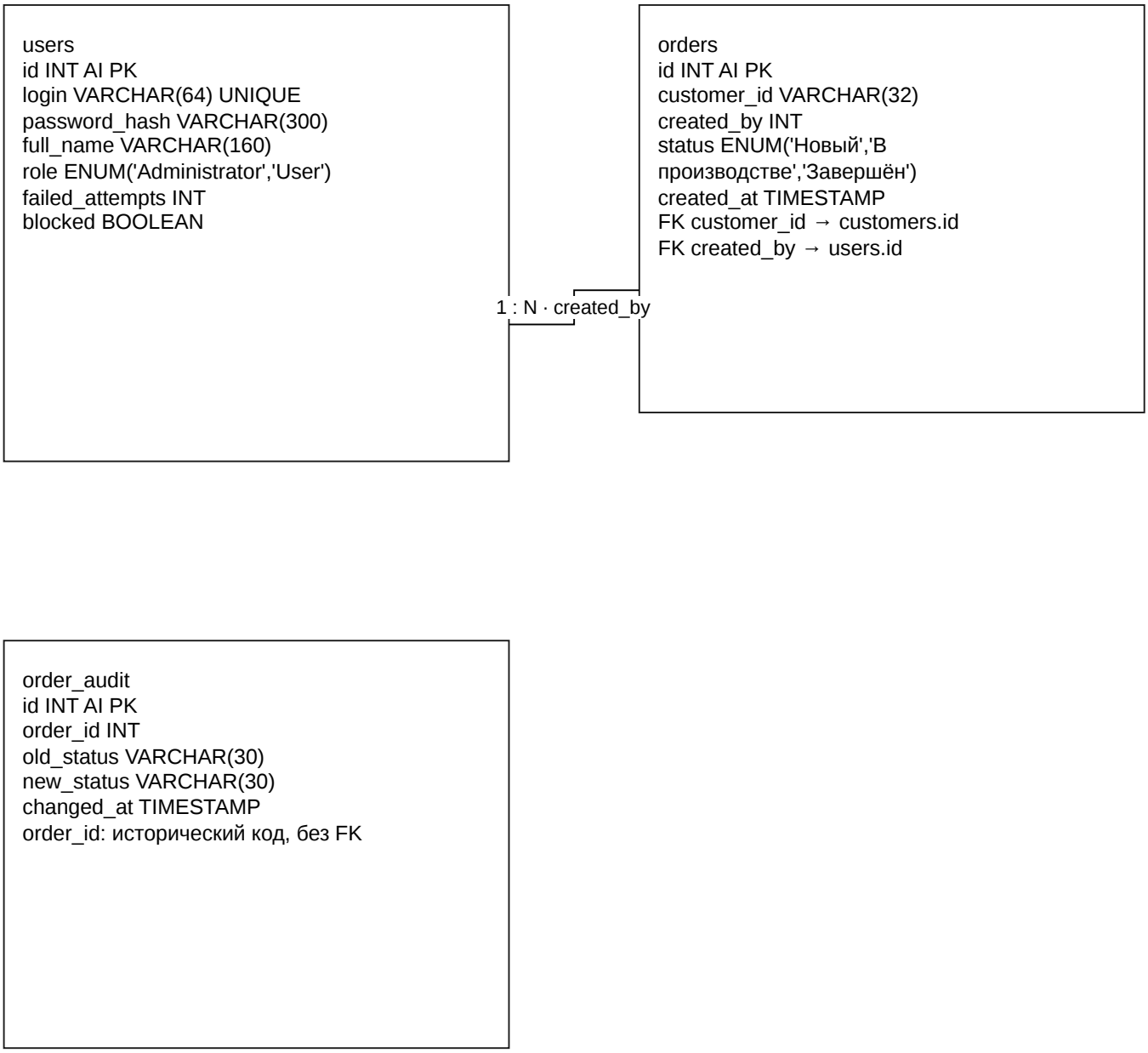

customers
id VARCHAR(32) PK
name VARCHAR(200)
inn VARCHAR(32)
address VARCHAR(300)
phone VARCHAR(80)
kind VARCHAR(80)
orders
id INT AI PK
customer_id VARCHAR(32)
created_by INT
status ENUM('Новый','В производстве','Завершён')
created_at TIMESTAMP
FK customer_id → customers.id
FK created_by → users.id
1 : N · customer_id
1 : N · order_id
products
id INT PK
code VARCHAR(40) UNIQUE
name VARCHAR(200)
order_lines
order_id INT
product_id INT
quantity INT
product_id)
PK (order_id,product_id)
FK order_id → orders.id
FK product_id → products.id
1 : N · product_id
1 : 0..1
production_orders
id INT AI PK
order_id INT UNIQUE
launch_date DATE
department VARCHAR(160)
FK order_id → orders.id


products
id INT PK
code VARCHAR(40) UNIQUE
name VARCHAR(200)
1 : N · product_id
1 : N · product_id
material_norms
product_id INT
material_id INT
norm DECIMAL(14,6)
material_id)
PK (product_id,material_id)
FK product_id → products.id
FK material_id → materials.id
operation_norms
product_id INT
operation_id INT
hours DECIMAL(14,6)
operation_id)
PK (product_id,operation_id)
FK product_id → products.id
FK operation_id → operations.id
1 : N · material_id
1 : N · operation_id
materials
id INT PK
name VARCHAR(200)
unit VARCHAR(30)
price DECIMAL(14,4)
operations
id INT PK
name VARCHAR(200)
rate_hour DECIMAL(14,4)


users
id INT AI PK
login VARCHAR(64) UNIQUE
password_hash VARCHAR(300)
full_name VARCHAR(160)
role ENUM('Administrator','User')
failed_attempts INT
blocked BOOLEAN
1 : N · created_by
orders
id INT AI PK
customer_id VARCHAR(32)
created_by INT
status ENUM('Новый','В производстве','Завершён')
created_at TIMESTAMP
FK customer_id → customers.id
FK created_by → users.id
order_audit
id INT AI PK
order_id INT
old_status VARCHAR(30)
new_status VARCHAR(30)
changed_at TIMESTAMP
order_id: исторический код, без FK